

Quality Inspection Report for Selective Laser Melting of 316L Stainless Steel Flat Washers

Report Number: QIR-SLM-316L-2023-045

Date of Inspection: October 26, 2023

Inspector: J. Martinez, Senior Quality Engineer

Customer Reference: P/O 8842-B, Flat Washers Lot #FW-316L-33

Applicable Standards: ASTM F138/F139, ISO 9001:2015, Internal Specification QS-AM-07

1.0 Executive Summary

This report documents the quality inspection for the manufacturing batch of 316L stainless steel flat washers produced via Selective Laser Melting (SLM). The inspection was conducted to verify conformance with specified dimensional, material, and process requirements. All 33 finished parts met the acceptance criteria, with no non-conformances identified. Process parameters and material usage were recorded per standard operating procedures, and all waste streams were managed in accordance with environmental protocols.

2.0 Inspection Scope and Objectives

The inspection covered the complete SLM manufacturing cycle for 316L stainless steel flat washers, including:

- Verification of raw material inputs and handling
- Monitoring of process parameters during water atomization and SLM operations
- Dimensional and visual inspection of finished parts
- Documentation of material recovery and waste disposal
- Energy consumption recording

The objective was to ensure that the manufacturing process adhered to established quality plans and that the final products met all specified requirements for subsequent use in high-corrosion resistance applications.

3.0 Inspection Procedures

3.1 Material Verification

All incoming materials were verified against certificates of analysis. The stainless steel feedstock for powder production was confirmed to be X2CrNiMo1712 grade, equivalent to 316L composition. Process water quality was tested for purity before use in atomization.

3.2 Process Monitoring

The SLM process was monitored in real-time using the equipment’s built-in sensors and external data loggers. Key parameters including laser power, chamber atmosphere, and build plate temperature were recorded at 5-minute intervals throughout the 13.38-hour processing cycle.

3.3 Dimensional Inspection

Finished washers were inspected using a coordinate measuring machine (CMM) with 0.001 mm accuracy. Critical dimensions including outer diameter, inner diameter, thickness, and flatness were measured against drawing specifications.

3.4 Visual Examination

All parts underwent 100% visual inspection under 10x magnification for surface defects, including porosity, cracks, and surface finish irregularities.

3.5 Material Balance Verification

Input and output materials were weighed using calibrated scales with 0.01 kg accuracy. Recovery and waste streams were quantified and documented.

4.0 Process Parameters

The following parameters were recorded during the manufacturing cycle:

Parameter	Value	Unit	Notes
SLM Processing Time	13.38	hours	Complete build cycle
SLM Nominal Power (Laser Active)	5.5	kW	During melting operations
SLM Nominal Power (Laser Inactive)	3.5	kW	During recoating and cooling phases
Argon Volume per Component	54	liters	Shielding gas during melting
Argon Volume for Chamber Filling	700	liters	Initial atmosphere establishment
Components per Build Cycle	33	pieces	Total parts in single build
WA Energy Consumption	2.23	MJ/kg	Historical average for melting operation

Note: WA energy consumption represents historical data for reference purposes.

5.0 Material and Energy Inputs

5.1 Material Inputs

The following materials were consumed during the manufacturing process:

Material Description	Quantity	Unit	Purpose
X2CrNiMo1712 Stainless Steel	4.11	kg	Powder production feedstock
Process Water	16.8	kg	Cooling medium for water atomization
Argon Gas	3.08	kg	Shielding and process atmosphere

5.2 Energy Inputs

Electrical energy consumption was recorded separately for different process stages:

Process Stage	Energy Consumption	Unit
Water Atomization	2.55	kWh
Selective Laser Melting	64.92	kWh

Historical Note: SLM energy consumption for similar batches has averaged approximately 65 kWh, showing consistent process stability.

6.0 Outputs and Waste Management

6.1 Product Output

The manufacturing cycle produced 33 finished flat washers meeting all specification requirements:

Product Description	Quantity	Total Mass	Unit
316L Stainless Steel Flat Washers	33	0.61	kg

6.2 Material Recovery

Efficient material recovery was implemented throughout the process:

Recovered Material	Quantity	Unit	Disposition
316L Powder Reused in SLM	2.94	kg	Returned to process hopper
316L Powder Returned for Remelting	0.15	kg	Sent to water atomization facility
Recovered Process Water	16.4	kg	Recycled to atomization system

6.3 Waste Streams

Non-recoverable materials were properly disposed of according to waste management protocols:

Waste Type	Quantity	Unit	Disposal Method
Solid Waste from Water Atomization	0.41	kg	Landfill
Non-recyclable 316L Powder	0.01	kg	Landfill

7.0 Quality Verification Results

7.1 Dimensional Inspection Results

All 33 washers were inspected against the critical dimensions specified in drawing DWG-316L-FW-04. The results showed 100% conformance:

Dimension	Specification	Average Measurement	Tolerance	Status
Outer Diameter	20.0 mm	19.998 mm	±0.05 mm	PASS
Inner Diameter	10.5 mm	10.502 mm	±0.05 mm	PASS
Thickness	2.0 mm	2.001 mm	±0.02 mm	PASS
Flatness	0.05 mm max	0.03 mm	N/A	PASS

7.2 Visual Inspection Results

Visual examination under magnification revealed no surface defects exceeding acceptable limits. All parts exhibited consistent surface finish with no visible porosity, cracks, or discoloration.

7.3 Material Composition Verification

Random sampling of three washers was subjected to X-ray fluorescence analysis, confirming 316L composition with chromium, nickel, and molybdenum content within specified ranges.

8.0 Process Efficiency Observations

The material utilization efficiency for this batch was calculated at approximately 85%, considering the ratio of finished product mass to total powder input. This aligns with established performance metrics for SLM processes with similar geometries.

Energy consumption per part averaged 2.06 kWh, which is consistent with historical data for components of this size and complexity. The argon consumption of 3.08 kg for the complete build cycle represents optimal usage within expected parameters.

9.0 Non-Conformances and Corrective Actions

No non-conformances were identified during this inspection. All process parameters remained within control limits, and finished parts met all specified requirements.

10.0 Conclusions and Compliance Statement

The manufacturing batch of 316L stainless steel flat washers produced via Selective Laser Melting has successfully passed all quality inspection criteria. The process was conducted in accordance with established procedures, and all material, energy, and waste data has been accurately documented.

Statement of Compliance: This batch of 33 flat washers conforms to all specified requirements and is approved for shipment to the customer.

Approvals:

J. Martinez

Senior Quality Engineer

Date: October 26, 2023

A. Chen

Production Manager

Date: October 26, 2023

Document Classification: Uncontrolled Copy when printed

Retention Period: 7 years from date of issue